EXPERIMENT - 15

AIM :

To implement a Decision Tree classifier using Python to predict whether a student passes or fails.o write a Python program to implement the A* search algorithm to find the least-cost path from a starting node to a goal node using path cost and heuristic estimation..

ALGORITHM:

1. Start the program.
2. Import the Decision Tree classifier.
3. Create a dataset containing study hours and attendance.
4. Assign pass or fail labels to the data.
5. Train the Decision Tree model.
6. Predict the result for a new student.
7. Display the prediction and accuracy.
8. Stop the program.

PROGRAM :

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```
# Decision Tree using ID3 Algorithm

import math
from collections import Counter

data = [
    ["Sunny", "No"],
    ["Sunny", "No"],
    ["Overcast", "Yes"],
    ["Rainy", "Yes"],
    ["Rainy", "Yes"],
    ["Rainy", "No"],
    ["Overcast", "Yes"],
    ["Sunny", "No"]
]

def entropy(rows):
    counts = Counter(row[-1] for row in rows)
    total = len(rows)
    result = 0

    for count in counts.values():
        p = count / total
        result -= p * math.log2(p)

    return result

def build_tree(rows, features):
    labels = [row[-1] for row in rows]

    if len(set(labels)) == 1:
        return labels[0]

    if not features:
        return Counter(labels).most_common(1)[0][0]

    best_feature = max(
        features,
        key=lambda f: entropy(rows) -
        sum(
            len([r for r in rows if r[f] == v]) / len(rows)
            * entropy([r for r in rows if r[f] == v])
            for v in set(r[f] for r in rows)
        )
    )
```

```
    tree = {best_feature: {}}

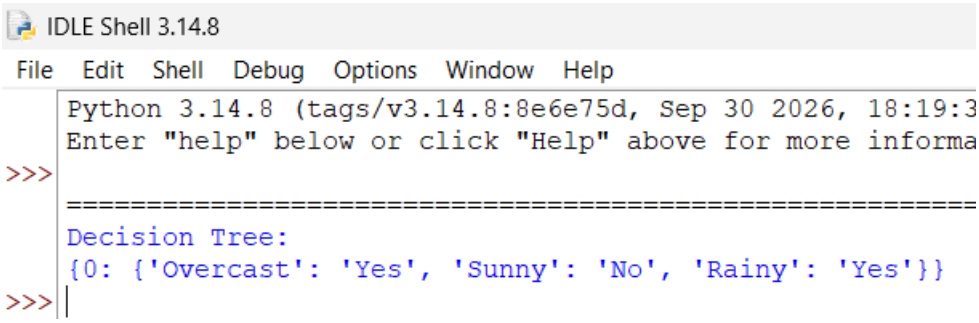
    for value in set(row[best_feature] for row in rows):
        subset = [r for r in rows if r[best_feature] == value]
        remaining = [f for f in features if f != best_feature]
        tree[best_feature][value] = build_tree(subset, remaining)

    return tree

features = [0]
tree = build_tree(data, features)

print("Decision Tree:")
print(tree)
```

OUTPUT:



RESULT:

Thus, the Decision Tree classifier was implemented successfully to predict the student's result based on study hours and attendance.